

Estimates, Uncertainty, and Statistical Tests

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Chapter overview

Chapter 2 introduced population and sample, parameter and statistic, sampling distribution and standard error. We now use those ideas to specify what a study seeks to estimate and to quantify uncertainty about the estimate obtained. A continuing comparison leads from an estimation rule to its standard error, a confidence interval, and a statistical test. We transfer the reasoning to scores measured twice on the same students and to categorical counts, with school comparisons developed in the reference appendix. We then bring magnitude, practical importance, error rates, and several comparisons together in a qualified conclusion.

A first-year course wanted to know whether a new practice format helps students remember what they have learned. Its instructors recruited 120 consenting students and randomly assigned 60 to Special Practice and 60 to Usual Practice. Both formats covered the same content over four sessions with equal total study time. Two weeks later, all 120 completed the same assessment, scored from 0 to 40. Special Practice averaged 27.1 points and Usual Practice 24.0, a difference of 3.1000 points. This study is invented for the chapter.

The difference is easy to calculate. Deciding what it tells us takes more work. A different assignment could have produced a different difference, even with the same participants. We need to understand that variation to judge how precisely this study estimates its intended comparison. We also need the outcome's scale and the course's purpose to judge whether a difference of this size matters. The calculations below develop those two judgments together.

1 From a Sample to an Estimate

1.1 The study and its estimated difference

The study asks about the **mean effect of assignment to Special rather than Usual Practice on the delayed assessment among these 120 participants**. Each part matters. Assignment is the intervention being compared; attending every session would be a different comparison. A mean effect summarizes the participants and allows individual responses to differ. The two-week assessment measures a specified outcome at a specified time. An effect on durable learning or among students elsewhere would require additional information.

The course team also chose a practical benchmark before seeing the outcomes. It treated two points as the smallest average difference that would change its recommendation: the marking scheme awards two points for one completed application task, and the new format adds little preparation time. This gives the benchmark a concrete, course-specific rationale. We will return to it after calculating uncertainty.

Table 1*Observed summaries for the invented special-practice study.*

Item	Special Practice	Usual Practice
Assigned students	60	60
Delayed assessments observed	60	60
Delayed-assessment mean	27.1	24.0
Delayed-assessment SD	6.0	6.4

The standard deviations describe differences among students' scores within each group. Scores vary by roughly six points around their group means, while the two means differ by 3.1000 points. That comparison of spreads is useful description, but the uncertainty of a *difference between means* is a separate quantity. Its calculation begins by naming the target and the rule used to estimate it.

1.2 Estimand, estimator, estimation, and estimate

Chapter 2 distinguished the population and sample, introduced parameter and statistic, and explained sampling distributions and standard error. We now use those ideas to distinguish the quantity sought, the estimation rule, and the estimate obtained.

An **estimand** is the precisely specified quantity the question seeks. For this study, it is the mean assignment effect just described. To make the target explicit, let $Y_i(1)$ be student i 's delayed score if assigned to Special Practice and $Y_i(0)$ the score if assigned to Usual Practice. These are the student's two **potential outcomes**. Only the one corresponding to the actual assignment is observed; Chapter 1 called the unobserved alternative the counterfactual outcome. The target is

$$\Delta\mu = \frac{1}{120} \sum_{i=1}^{120} \{Y_i(1) - Y_i(0)\}.$$

Inside the braces is one student's assignment effect. The sum adds all 120 effects, and division by 120 takes their mean. Written this way, $\Delta\mu$ is a mean difference over these 120 participants, not a population quantity. The notation allows different students to have different effects. We assume the two formats have a consistent meaning and one student's assignment does not change another student's outcome in a relevant way.

An **estimator** is a rule for using observed data to approximate this target. Here the rule subtracts the Usual assigned-group mean from the Special assigned-group mean. **Estimation** is the process of applying and checking that rule and characterizing its uncertainty. The **estimate** is the numerical value obtained:

$$\widehat{\Delta\mu} = \bar{y}_1 - \bar{y}_0 = 27.1 - 24.0 = 3.1000 \text{ points.}$$

The bar denotes a group mean, and the hat marks an estimate. The rule would stay the same under another admissible assignment, although the observed outcomes and the two group means could change. Reversing the subtraction gives -3.1000 ; it expresses the reverse comparison using the same information.

The observed difference exactly describes the two realized means. That description is known. The participant mean assignment effect remains unknown because no student reveals both potential outcomes. This is why an exact piece of arithmetic can still be an uncertain estimate.

Table 2

Inferential objects in the special-practice study.

Object	Working definition	In this study	Common reading error
Estimand	The precisely specified quantity the question seeks	Mean delayed-score difference caused by Special rather than Usual assignment among the 120 randomized participants	Naming only “the treatment effect”
Estimator	The rule that would be applied in each admissible repetition	Special-group mean minus Usual-group mean	Treating the rule as the observed result
Estimation	Applying and checking the estimator for the specified estimand and characterizing the estimate and its uncertainty	Apply the mean-difference rule, verify its inputs and conditions, and quantify uncertainty	Treating estimation as another number
Estimate	The number the estimator produced in this study	$27.1 - 24.0 = 3.1000$ points	Reporting a number without units or direction
Parameter	The fixed feature named by the estimand under the chosen population or process interpretation	The participant-specific mean assignment contrast	Assuming it is the sample estimate or a universal truth
Statistic	A numerical function of observed data	A group mean, the 3.1000-point estimate, or the 2.7372 test statistic	Assuming every statistic estimates the same thing

Object	Working definition	In this study	Common reading error
Standard error	The modeled spread of an estimator across the declared repetitions	1.1325 points for the mean-difference estimator	Calling it variation among students

The table distinguishes objects by their function. A statistic is any numerical function of the observed data; a group mean, an estimated SE, and a test statistic all qualify. A parameter is fixed within the specified population or process interpretation. Calling it fixed does not make it universal: changing the participant set, outcome, time, or condition comparison may change the estimand and its parameter.

1.3 Name the repetition and working model

Uncertainty needs a description of what could have varied. In a sampling question, different people might have been selected from a population. In the special-practice study, the primary target concerns these same 120 participants. The relevant repetition assigns exactly 60 of them to each format again. Their potential outcomes remain fixed; the assigned groups and observed outcomes can change.

Random assignment makes the difference of assigned-group means useful for estimating the participant effect. It does not turn these consenting students into a representative sample of all first-year students. Inference about a wider population would need a stated model or sampling justification and an explicit extension of the target.

The numerical analysis uses the familiar **Welch procedure** as an approximate working calibration (Welch, 1947). It allows different observed group variances. Its formulas also occur in independent-sample comparisons, but complete assignment of 60 out of 120 students does not make the two assigned-group means independent. Sections 2 and 3 explain what the formula estimates here and what its approximation requires. Appendix A gives the design-based variance identity and the distinction between an average-effect null and a sharp null.

1.4 Expectation, bias, and precision

Imagine applying the estimator over all repetitions allowed by the design or model. Its expectation, in Chapter 2's sense, is the probability-weighted average of the values it would produce across those repetitions. If an estimator averages 3.4 points when the target is 3.0, its bias is $3.4 - 3.0 = 0.4000$ points. This describes the rule's systematic displacement, not the error of one particular result.

Under complete random assignment and the stated potential-outcome conditions, each assigned-group mean is centered on the corresponding participant mean potential outcome. Their difference is therefore unbiased for the participant mean assignment effect:

$$\text{Bias}(\bar{Y}_1 - \bar{Y}_0) = E_{\text{assign}}(\bar{Y}_1 - \bar{Y}_0) - \Delta\mu = 0.$$

The equality does not say that this estimate, 3.1000, equals the unknown effect. **Precision** concerns how tightly the estimator's values cluster across assignments:

$$SE(\bar{Y}_1 - \bar{Y}_0) = SD(\bar{Y}_1 - \bar{Y}_0 \text{ across possible assignments}).$$

An unbiased rule can be imprecise; a tightly concentrated rule can be biased. A third useful consideration is how its behavior changes under unusual observations or departures from its conditions. A small SE under an unsuitable model can give unwarranted confidence. Appendix A develops the bias-variance trade-off using mean squared error.

2 Using Sampling Variation to Quantify Uncertainty

2.1 From the visible sampling pattern to an SE formula

Chapter 2 Section 4 generated 5,000 independent waiting-time samples at each of three sample sizes. The waiting-time model had mean and SD both equal to 10 minutes. The sample means varied less when a sample contained more observations. Increasing the number of simulated repetitions made their distribution easier to see; increasing the size of each sample changed the distribution's spread.

We can now explain that spread algebraically. Suppose $Y_1, \dots, Y_n$ are independent observations from the same distribution, with finite variance σ^2 . The sum has variance $n\sigma^2$, because independent variances add. The mean divides that sum by n ; multiplying a variable by $1/n$ multiplies its variance by $1/n^2$. Thus

$$\text{Var}(\bar{Y}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}, \quad SE(\bar{Y}) = \frac{\sigma}{\sqrt{n}}, \quad \widehat{SE}(\bar{Y}) = \frac{s}{\sqrt{n}}.$$

The last expression estimates the unknown population SD with the sample SD. The estimated SE is calculated from one sample; the SE it estimates describes the mean across repetitions. Both are standard deviations, but of different distributions from the individual waiting times.

For the waiting-time model, the theoretical SEs are $10/\sqrt{5} = 4.4721$, $10/\sqrt{30} = 1.8257$, and $10/\sqrt{100} = 1.0000$ minutes. Chapter 2's empirical values, about 4.54, 1.79, and 1.01, approximate those targets. Simulation noise explains small discrepancies. The square-root

relationship also explains why halving an SE usually requires four times as many independent observations under the same model.

2.2 Uncertainty for a difference in means

For two independent samples, uncertainty comes from both means. Their variances add when one mean is subtracted from the other:

$$\text{Var}(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}, \quad \widehat{SE}(\bar{y}_1 - \bar{y}_0) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

The minus sign changes the direction of a difference, not the sign of a variance contribution. Independence removes the covariance term; the next subsection shows what happens when that condition fails.

Return now to the fixed-participant assignment study. Its estimated variance has the same numerical form, with a different design-based interpretation. Substitution gives

$$\begin{aligned} \widehat{SE}(\bar{y}_1 - \bar{y}_0) &= \sqrt{\frac{6.0^2}{60} + \frac{6.4^2}{60}} = \sqrt{\frac{36}{60} + \frac{40.96}{60}} \\ &= \sqrt{0.6000 + 0.682666 \dots} = 1.1325 \text{ points.} \end{aligned}$$

Each variance is in squared assessment points. Division by a group size gives its contribution to uncertainty, and the final square root returns to points. Keep unrounded quantities while calculating and report 1.1325 points in the chapter’s result.

Across assignments of these participants, the exact variance contains a subtraction for variability in individual effects. We cannot recover that variability, because no student reveals both potential outcomes. The squared SE above omits this subtraction and estimates the remaining terms. It estimates an upper bound on assignment variance *on average*, with equality when individual effects are constant. This does not mean every realized SE is too large. Appendix A shows the identity that supports this statement (Bai et al., 2024).

The working reference also needs adequate information in each group and no dominating observations. Random assignment alone does not make the Welch reference exact. Figure 1 illustrates the spread using 500 Normal draws centered at 3.1 with SE 1.1325. The center is set to the observed value to make the display concrete; it is not a claim that the unknown estimand equals 3.1. These are simulated values of the same estimator, not additional study evidence or the enumerated randomization distribution.

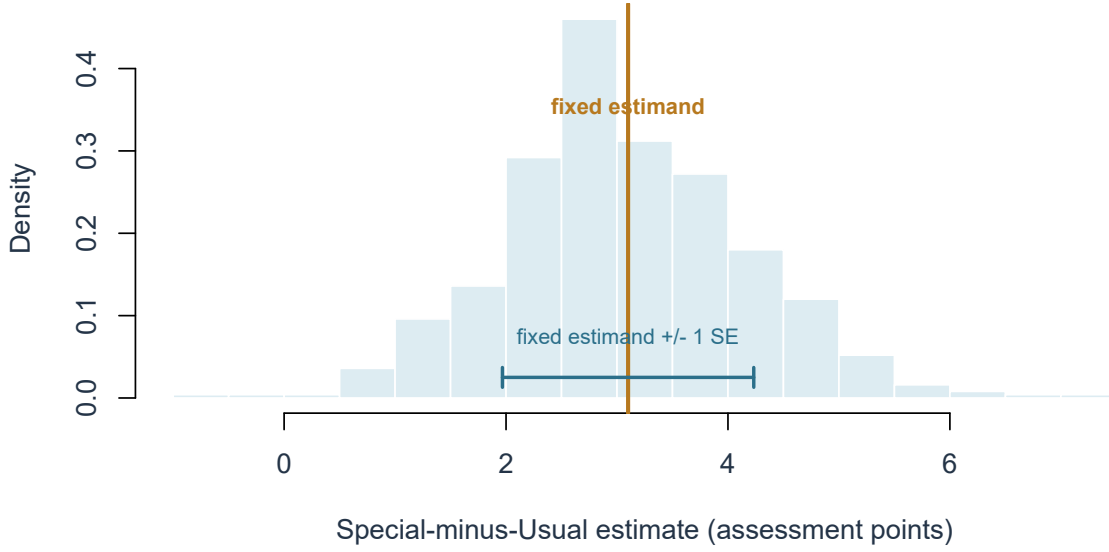


Figure 1: An illustrative simulation of repeated Special-minus-Usual estimates. The estimand and estimator remain fixed; realized estimates vary, and their modeled spread is the standard error.

2.3 Pairing, clustering, and sampling without replacement

Dependence changes a variance calculation. For any two quantities with finite variances,

$$\text{Var}(U - V) = \text{Var}(U) + \text{Var}(V) - 2 \text{Cov}(U, V).$$

If paired measurements tend to be high or low together, subtracting within a pair removes some shared variation. In a separate illustration, suppose both measurements have SD 4, their covariance is 12, and there are 25 independent pairs. Let d be the within-pair difference. Then

$$\begin{aligned} \text{Var}(d) &= 4^2 + 4^2 - 2(12) = 8.0000, \\ \widehat{SE}(\bar{d}) &= \sqrt{\frac{8}{25}} = 0.5657, \\ \text{ignoring the pairing: } &\sqrt{\frac{16 + 16}{25}} = 1.1314. \end{aligned}$$

The reduction comes from the positive covariance, not from a change in the number of observations. Actual paired-data analysis computes each difference and uses the SD of those differences. Negative covariance would change the result in the opposite direction.

An invented paired-school feedback illustration provides a different form of dependence. It measured 480 students in 24 classes within 12 schools, assigning one class in each school to each condition. Its uncertainty follows the 12 within-school contrasts, which Appendix B works through. Counting 480 distinct students does not establish 480 independent assignments.

How much information data provide depends on the estimand, dependence, weighting, and design. Repeated rows for the same person can add less information than equally many independent observations. In simple random sampling without replacement, observing a large fraction of a finite population leaves less uncertainty about its mean. Appendix A gives the finite-population correction and explains a precision-equivalent effective sample size under a stated comparison model.

2.4 Concentration, shape, and the conditions behind them

Two ideas explain different features of Chapter 2's repeated means. The **law of large numbers** describes concentration: under its conditions, a sample mean gets close to the population mean as sample size grows. The **central limit theorem** describes shape: a suitably centered and scaled sample mean approaches a Normal distribution under conditions such as independent, identically distributed observations with finite variance. The original waiting times can remain strongly skewed.

These results concern increasing sample size, not repeatedly simulating the same small sample. They also require conditions. Strong dependence, extreme tails, or an observation that dominates the mean can make a familiar Normal approximation poor. There is no universal sample-size cutoff that settles the issue for every distribution and design.

A running mean makes the concentration claim visible. In Figure 2, each point averages all the exponential waiting times accumulated so far. The population mean stays at ten minutes while the sample mean changes as observations arrive.

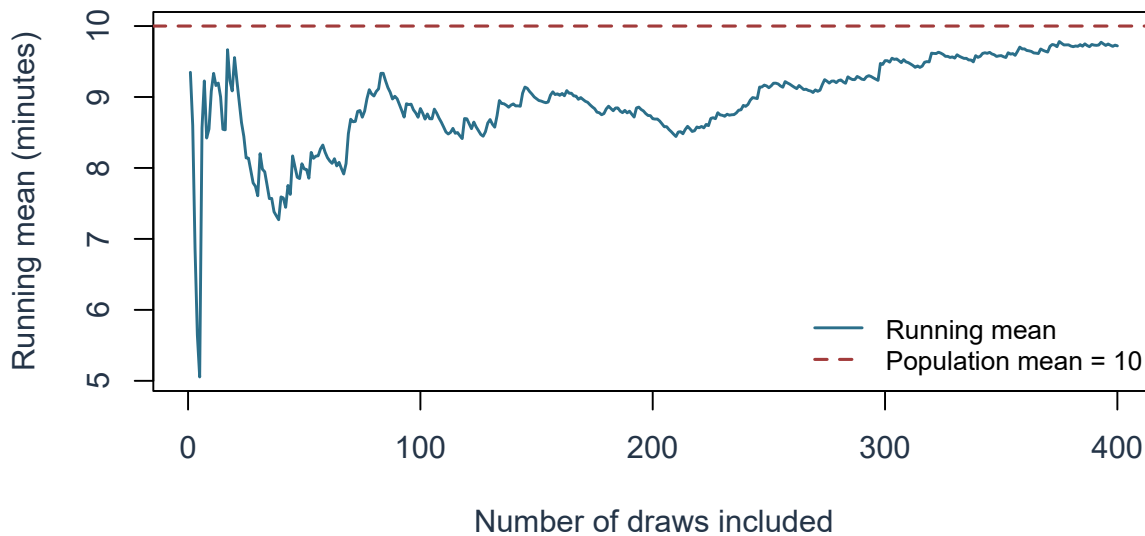


Figure 2: A running mean from 400 independent exponential waiting times with population mean ten minutes.

The running mean finishes at 9.7207 minutes. Its early shortfall does not change the population mean, and the next observation can move the average toward or away from ten. The individual waiting times remain skewed. This is one accumulating sequence; to examine the CLT's shape claim, we need a distribution of means across repeated samples of the same size, as in Chapter 2.

The SE formula answers how wide a sampling distribution is under a model. An approximate shape then helps choose an interval or test. Keep those two jobs distinct: a correct spread calculation alone supplies no tail probability. Appendix A gives the convergence notation and fuller conditions.

3 Confidence Intervals

3.1 Estimate, SE, critical value, and interval width

A confidence interval combines an estimate with a multiple of its SE. For the symmetric procedures used here,

$$\text{estimate} \pm \text{critical value} \times \text{standard error}.$$

The SE supplies a scale. A **critical value** says how far to extend in SE units to obtain the intended coverage under a reference distribution. Chapter 2 introduced the standard Normal

distribution, centered at zero with SD one. Its .975 quantile is about 1.9600: .025 of the probability lies beyond that point, and another .025 lies below -1.9600 .

A **t distribution** is also symmetric around zero but has heavier tails, especially with few degrees of freedom. In a one-mean procedure, these tails account for estimating the spread. As degrees of freedom increase, the t reference approaches the standard Normal. An estimated spread does not make every statistic exactly t-distributed: the exact one-mean result requires independent Normal sampling, while Welch's reference is approximate.

The 95% reference values below leave .025 in each tail. At low degrees of freedom, estimating spread produces a larger t critical value; the larger-df reference is closer to Normal.

Table 3

Reference choice and critical values.

Reference	95% critical value	Role in the examples
Standard Normal	1.9600	Mean with a population SD known independently of the sample, under a suitable Normal model
t with 24 df	2.0639	One mean with SD estimated from 25 independent Normal observations
t with 117.5119 df	1.9804	Approximate Welch reference for the supplied Special Practice summaries

The procedure and its conditions determine the reference. Choosing the smaller critical value merely to narrow an interval would change the procedure without justification.

Chapter 2 used a **z score**, $z_i = (x_i - \mu)/\sigma$, to locate one observation in population SD units. Sample standardization, $(x_i - \bar{x})/s$, describes an observation relative to its sample. Neither operation makes the original distribution Normal. For a mean, the quantity being standardized is instead its estimator, so the denominator is an SE:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}, \quad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1.$$

The **one-sample z test** uses a population SD known independently of the sample. The **one-sample t test** estimates that SD with s . Its exact t reference requires independent observations from a Normal population; suitable distributional conditions and enough information can support an approximation more broadly. A large sample does not turn an unknown SD into a known one or remove dependence.

Figure 3 compares references for the same standardized statistic. With a statistic of 2.0000, the two-sided probabilities are .0455 for Normal, .0766 for t with 9 df, and .0482 for t with 99 df. The change comes from the reference tails. Changing a study's sample size can also change its SE and observed statistic, which is a separate comparison.

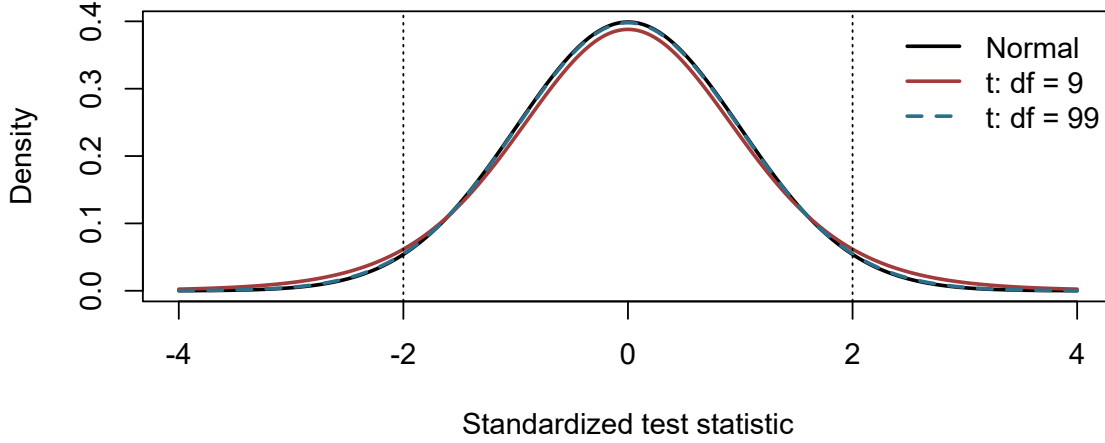


Figure 3: Normal and t references at the same location and scale of the test statistic. The heavier t tails are most visible with few degrees of freedom.

For the special-practice procedure, the Welch degrees of freedom are 117.5119 and the 95% critical value is 1.9804. The fractional degrees of freedom describe this reference approximation; they are not a count of students. Appendix A provides the full formula. The interval is

$$\begin{aligned}
 (\bar{y}_1 - \bar{y}_0) \pm t_{.975, 117.5119} \widehat{SE}(\bar{y}_1 - \bar{y}_0) &\approx 3.1000 \pm 1.9804(1.1325) \\
 &\approx 3.1000 \pm 2.2429 \\
 &\approx [0.8571, 5.3429] \text{ points.}
 \end{aligned}$$

The endpoints retain the Special-minus-Usual direction and assessment-point units. They allow positive effects near one point and effects above five points. The data therefore give a useful estimate without pinpointing its magnitude.

For a separate illustration, suppose observations are independently sampled from the same Normal population, with $n = 25$, $\bar{x} = 52$, and $s = 5$. To estimate its population mean, use $\widehat{SE}(\bar{X}) = 5/\sqrt{25} = 1.0000$ point. With 24 degrees of freedom and the supplied critical value 2.064,

$$52 \pm 2.064(1.0000) = [49.9360, 54.0640].$$

The mechanics are the same, but the estimand and repetition have changed from a participant assignment effect to a population mean under sampling.

3.2 Coverage belongs to the repeated procedure

A 95% confidence procedure aims to cover the fixed estimand in 95% of its repeated applications under its conditions. Across repetitions, the observations, estimate, and endpoints vary; the target stays fixed.

Figure 4 shows 40 applications of a simplified Normal procedure with a known SE. The estimate is drawn from that model, and the interval adds and subtracts $1.96SE$. Most intervals cross the fixed target line; a few miss. A batch of 40 need not contain exactly 38 successes even when the procedure has 95% coverage. The figure's coverage describes this simplified model.

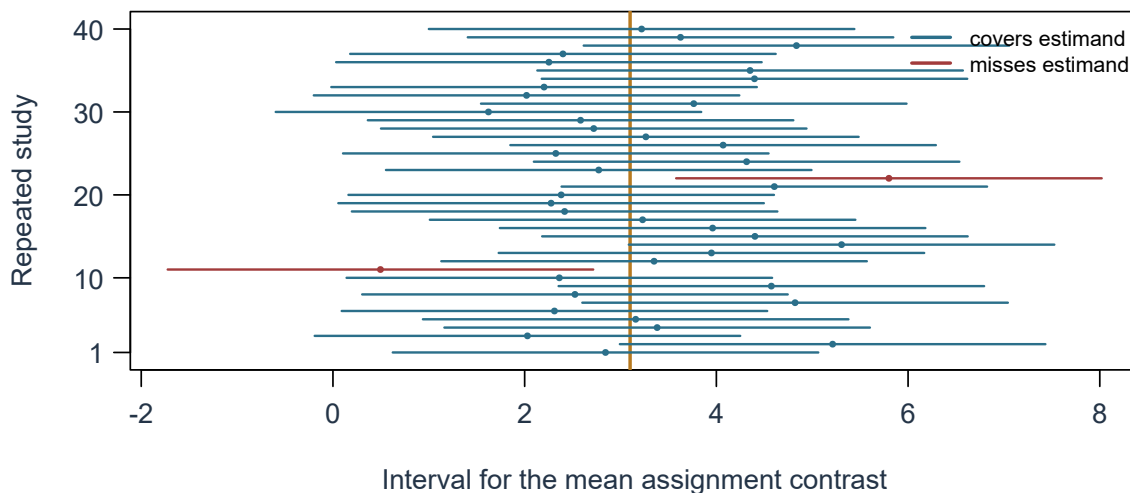


Figure 4: Forty intervals from one calibrated illustrative procedure. The estimand remains fixed while estimates and intervals vary; occasional misses are compatible with 95% long-run coverage.

Once the actual interval has been calculated, the frequentist coverage statement does not assign a 95% probability to its fixed estimand. Neither does it say that 95% of students score between 0.8571 and 5.3429, or that 95% of repeated estimates fall inside this particular interval. It describes the success rate of the *interval-producing procedure*.

For interpretation, read the endpoints as values compatible with the observations under the procedure at the stated level. This does not make all values inside equally plausible or values just outside impossible. The interval expresses a range of uncertainty, with no natural cliff at its rounded endpoints.

3.3 What a narrower interval can and cannot resolve

Holding the model and data fixed, a higher confidence level needs a wider interval. More outcome variability generally widens it, while more relevant information narrows it. Changes in the design or variance estimator can also change its width. A useful target width depends on which differences would alter a substantive conclusion.

For the Special Practice comparison, changing only the confidence level gives the following intervals. Both use the same 3.1000-point estimate, unrounded SE, degrees of freedom, and observations.

Table 4

Changing confidence while keeping the estimate and SE fixed.

Confidence level	Critical value	Margin	Interval in points	Full width
90%	1.6579	1.8777	[1.2223, 4.9777]	3.7554
95%	1.9804	2.2429	[0.8571, 5.3429]	4.4857

Predict the changes before comparing the rows: the smaller critical value narrows the 90% interval without adding observations or reducing its SE. The narrower interval has a lower coverage target. Both intervals lie above zero and span the two-point benchmark.

The special-practice interval lies above zero but crosses the two-point benchmark. It can support a positive mean assignment effect under the procedure while leaving practical importance unresolved. We develop that joint reading in Section 5.

i An interval describes particular uncertainty

The interval quantifies the uncertainty represented by its procedure. A narrower interval can coexist with systematic selection, poor measurement, unmodeled dependence, unavailable outcomes, or an unsupported generalization. Those problems need information about the study and suitable analysis, in addition to precision.

For example, measuring the wrong outcome very consistently can yield a small SE for the wrong target. Adding many pupils from a few jointly assigned classes does not reproduce the information from the same number of independent assignments. Precision helps only when the procedure remains aligned with the question and design.

4 Statistical Tests and Worked Comparisons

4.1 Hypotheses, test statistics, reference distributions, and tail areas

A statistical hypothesis makes a proposition about a target. Begin with the separate one-mean example: $H_0 : \mu = 50$ says the population mean equals 50 points, and $H_A : \mu \neq 50$ allows either direction. The sample has $n = 25$, $\bar{x} = 52$ and $s = 5$. Its statistic is

$$t = \frac{52 - 50}{5/\sqrt{25}} = 2.0000, \quad df = 24, \quad p = 2P(T_{24} \geq 2) = .0569.$$

The numerator is the observed departure from the null mean. The denominator is the estimated SE of the mean, so the departure is two estimated SEs. The t reference with 24 df supplies the probability in both tails. Equivalently, subtracting 50 from every score creates a centered variable whose null mean is zero; the question stays the same.

For two independent populations, write $H_0 : \Delta\mu = 0$, where $\Delta\mu = \mu_1 - \mu_0$ and the subscripts identify the two groups. For Special Practice, $\Delta\mu$ is the fixed participants' mean assignment effect defined in Section 1.2, and its null preserves that target:

$$H_0 : \Delta\mu = 0 \quad \text{versus} \quad H_A : \Delta\mu \neq 0.$$

The null says the participant mean assignment effect is zero; the alternative allows either a positive or a negative effect. A test is the procedure for assessing this proposition, while a test result is the numerical output obtained from the data. A common statistic is

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error under the test procedure}}.$$

The numerator measures departure in outcome units. The denominator expresses how large that departure is relative to the procedure's uncertainty. Here

$$t = \frac{\widehat{\Delta\mu} - 0}{\widehat{SE}(\bar{y}_1 - \bar{y}_0)} \approx 2.7372, \quad p = 2P(T_{117.5119} \geq |t_{\text{obs}}|) \approx .0072.$$

The estimate is about 2.7372 estimated SEs above zero. The symbol $T_{117.5119}$ denotes a variable with the reference t distribution. Using the unrounded statistic and degrees of freedom, its upper-tail probability is .0036. The two-sided test adds the equally extreme negative tail, giving .0072. Software or a reference table supplies the area under the approximate Welch working procedure.

A **p-value** is this conditional tail probability. If the zero-effect null and the procedure's conditions hold, a result at least this incompatible would occur with probability about .0072. The calculation starts by assuming the null for its reference; it does not calculate the probability that the null itself is true (Wasserstein & Lazar, 2016).

A **significance level**, α , is an advance threshold for a decision rule. A level-.05 rule rejects when its calibrated p-value is below .05; .0072 meets that rule. Calling the result statistically significant records this comparison. It supplies neither the size of the effect nor a study-specific probability of having made an error.

The focal test is two-sided because either direction counts. A one-sided test needs an alternative whose direction was chosen before the result was inspected. A large effect in the opposite direction does not support that alternative. Selecting whichever tail favors the observed estimate changes the testing procedure and its error calibration.

Different t procedures use different units and variance calculations. Their shared name does not make paired and independent data interchangeable.

Table 5

Named t tests and the observations that carry information.

Procedure	Data and null	SE and reference
One-sample t	One quantitative sample; $H_0 : \mu = \mu_{\text{null}}$	$s/\sqrt{n}$; $df = n - 1$
Welch independent-samples t	Two independent samples; $H_0 : \mu_1 - \mu_0 = 0$	$\sqrt{s_1^2/n_1 + s_0^2/n_0}$; Welch-Satterthwaite df
Pooled independent-samples t	Two independent samples with a common population variance; same mean null	$s_p\sqrt{1/n_1 + 1/n_0}$; $df = n_1 + n_0 - 2$
Paired-samples t	Correctly matched observations, independent pairs; $H_0 : \mu_D = 0$	$s_D/\sqrt{n}$ on pair differences; $df = n - 1$

The **pooled t test** combines the two within-group variance estimates:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2}{n_1 + n_0 - 2}.$$

It is the **equal-variance t test**: it assumes a common population variance, uses $n_1 + n_0 - 2$ df, and when that assumption holds it is the more powerful of the two procedures, with an exact t reference and a slightly narrower interval for the same data. Welch's **unequal-variances t test** allows the group variances to differ; its estimated df are smaller, so it gives up a little power in exchange for protection when the variances are unequal, and it is the safer default

when a common variance cannot be justified from the design. For the equal-size Special Practice groups the two statistics coincide at 2.7372; only the df differ (118 versus 117.5119), and both p-values display as .0072. Equal sample sizes do not establish equal variances, and choosing a procedure solely from a preliminary variance-test p-value obscures the design and model; state the choice with its reason. Pairing comes from how observations were obtained; its t procedure analyzes within-pair differences. Chapter 4's term *pairwise comparisons* refers to several group contrasts, a different use of the word.

Table 6

Six mistranslations of the displayed $p = .0072$ and their repairs.

Mistranslation	Why it fails	Repair
"There is a 0.72% chance the null is true."	The p-value conditions on the null; it does not assign probability to it	Under the zero-contrast null and procedure, results at least this incompatible have probability .0072
"There is a 0.72% chance the result happened by chance."	Chance is not a complete model; the result already happened	Name the null, design, statistic, and reference process
"The study has a 99.28% chance of replicating."	Replication depends on the effect, design, sample, and analysis	Report the current estimate and uncertainty; discuss replication separately
"The effect is large or important because $p = .0072$."	A p-value is not an effect-size or importance measure	Interpret 3.1000 points and its interval against a defensible benchmark
"The study's false-positive probability is 0.72%."	A long-run error rule and a data-dependent p-value differ	State any prespecified decision rule; avoid a posterior error claim
"The p-value keeps its usual meaning after this outcome or model was selected from inspected results."	The selection rule is part of the procedure and can alter calibration	Report the family of analyses and whether the comparison was chosen in advance; interpret the selected result accordingly

The repairs in the table retain what the test can answer while directing other questions to the information they require. Replication needs a new-study model; practical importance needs an outcome scale and a benchmark. Reporting the estimate and interval alongside the p-value makes these distinctions easier to maintain.

4.2 A matched interval and test

A two-sided test at the 5% level built from the same procedure as a 95% interval agrees with that interval about zero. Here the estimator, SE, reference distribution, sidedness, and comparison all agree. The test rejects zero precisely when the interval excludes it. Figure 5 places the tail area and interval beside one another.

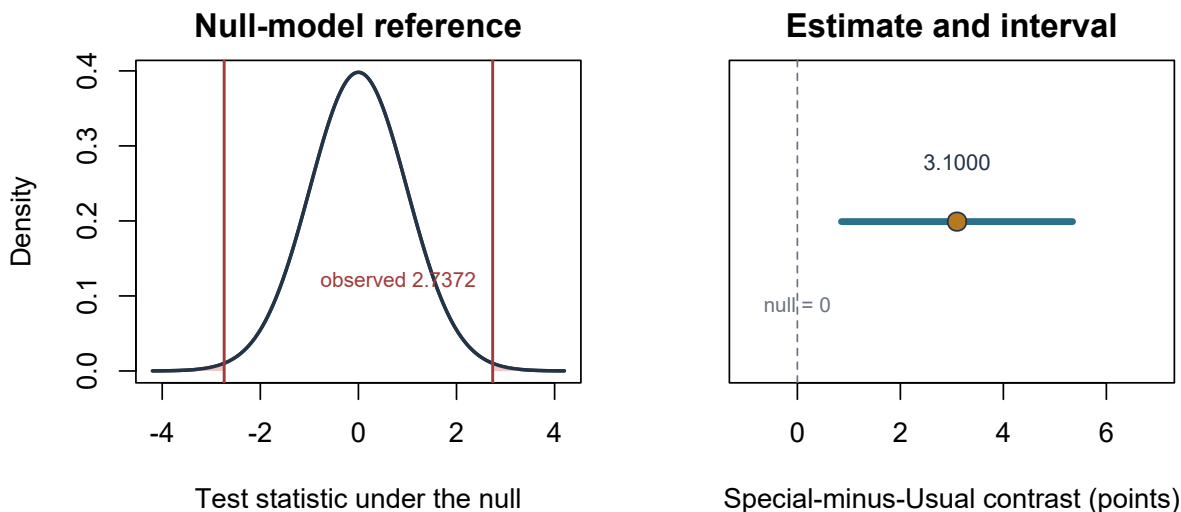


Figure 5: The two-sided test and 95% interval for the same contrast, built from the same procedure. The null value, standard error, and model remain aligned.

Algebra explains the agreement: zero lies outside $\widehat{\Delta\mu} \pm t^* \widehat{SE}$ when $|\widehat{\Delta\mu}|/\widehat{SE} > t^*$. A different SE, tail rule, reference, or multiplicity adjustment can break the correspondence. Two mechanically aligned calculations can also share an inadequate model. Agreement checks their construction; it does not validate their conditions.

For the separate one-mean example, testing $\mu = 50$ gives $t = (52 - 50)/1 = 2.0000$. This is smaller than the supplied two-sided critical value 2.064, and the interval $[49.9360, 54.0640]$ includes 50. The procedure therefore fails to reject 50 at the 5% level. That outcome does not establish that the mean equals 50: other values remain compatible as well.

4.3 Paired-samples and independent-samples t tests

Eight invented students completed the same-score-scale assessment before and after a short practice session. These are separate teaching records from Special Practice. Define each student's difference as $d_i = \text{after}_i - \text{before}_i$, preserving the person and subtraction direction before calculating a summary.

Table 7

Invented paired student scores; these are not Special Practice records.

Student	Before	After	Difference
1	50	49	-1
2	54	54	0
3	48	49	1
4	60	62	2
5	55	57	2
6	58	61	3
7	62	66	4
8	53	58	5

The differences sum to 16, so their mean is $16/8 = 2.0000$ points. Their squared deviations from two sum to 28. The SD therefore uses seven degrees of freedom:

$$\bar{d} = 2.0000, \quad s_d = \sqrt{28/7} = 2.0000, \quad \widehat{SE}(\bar{d}) = 2/\sqrt{8} = 0.7071.$$

Under an independent-pair model, the population mean change is μ_D . Testing a zero mean change and constructing the corresponding 95% interval gives

$$H_0 : \mu_D = 0, \quad t = \frac{2.0000 - 0}{2/\sqrt{8}} = 2.8284, \quad df = 7, \quad p = .0255,$$

$$\bar{d} \pm t_{.975,7} \frac{s_d}{\sqrt{8}} \approx 2.0000 \pm 2.3646(0.7071) \approx [0.3280, 3.6720] \text{ points.}$$

The calculation retains unrounded values. An exact t reference requires independent Normal pair differences; with other distributions its approximation needs justification. The result describes mean change under the stated sampling model. Without an untreated comparison, the session cannot be separated from time trends, repeated testing or other concurrent changes.

The variance identity in Section 2.3 explains why pairing matters:

$$\text{Var}(Y_{\text{after}} - Y_{\text{before}}) = \text{Var}(Y_{\text{after}}) + \text{Var}(Y_{\text{before}}) - 2 \text{Cov}(Y_{\text{after}}, Y_{\text{before}}).$$

Treating the eight before scores and eight after scores as independent groups would impose an inappropriate independence structure on these paired records and omit their within-student covariance. Its p-value cannot justify that change. For genuinely independent groups, Welch or the pooled procedure instead follows the group variances and their conditions.

The school-pair example in Appendix B extends this logic to class means. It prints all class means and works through the calculation: twelve within-school contrasts, rather than 480 individual scores, carry the assignment uncertainty. Chapter 4 recalls the purpose of blocking.

4.4 Transfer from means to categorical counts

The same question–estimate–reference sequence can assess a categorical comparison. Consider an invented teaching model with 100 independently sampled learners. Participation format and completion are recorded for each learner. The format was observed rather than randomly assigned, so the question concerns association.

Table 8

Invented counts for the first Pearson chi-square calculation.

Participation format	Completed	Did not complete	Total
Online	32	18	50
In person	18	32	50
Total	50	50	100

Among the 50 Online learners, 32 completed: $32/50 = .6400$. Among the 50 In-person learners, 18 completed: $18/50 = .3600$. The observed Online-minus-In-person difference is .2800, or 28.0000 percentage points. Each denominator is its own format’s total.

The null says format and completion are independent in the stipulated population or process. Under that restriction both formats have a common completion probability. Estimating it from all records gives $50/100 = .5000$. A cell’s **expected count** equals its row total times column total divided by the total number of records:

$$E_{rc} = \frac{(\text{row total})(\text{column total})}{n} = \frac{50(50)}{100} = 25.$$

Every cell has expected count 25 here. This pattern preserves the observed margins while imposing the null relationship. An expected count is a model-based reference; in other tables it can be fractional.

Table 9*Expected counts and contributions for the invented table.*

Cell	Observed O	Expected E	$(O - E)^2 / E$
Online, completed	32	25	1.9600
Online, did not complete	18	25	1.9600
In person, completed	18	25	1.9600
In person, did not complete	32	25	1.9600

Pearson's statistic adds squared observed-minus-expected departures, each scaled by its expected count:

$$X^2 = \sum_{r,c} \frac{(O_{rc} - E_{rc})^2}{E_{rc}} = 4(1.9600) = 7.8400.$$

With the margins accounted for, specifying one cell in a two-by-two table fixes the remaining three. Thus $df = (2 - 1)(2 - 1) = 1$. The nonnegative squared discrepancies become larger as observations depart from the null pattern, so the test uses the upper tail of a chi-square reference. Its probability is $p = .0051$.

Under the stipulated model, the table gives evidence against independence. The proportions give the direction and magnitude that the squared statistic discards: completion is 28.0000 percentage points more frequent in the Online group. This observed association does not establish that assigning a format would change completion.

The approximation requires independent observational units, mutually exclusive cell coding and adequate expected counts. Expected counts of 25 do not themselves establish independent sampling or a causal design. Chapter 6 reproduces the uncorrected Pearson calculation in R; software defaults can use a continuity correction for two-by-two tables, producing a related but different calculation. Exercise 7 supplies a separate 200-record table for transferring this reasoning.

The UCI intention-by-pass table in Chapter 2 has a different source, population and denominator. Chapter 6's categorical reference develops its inference and coding sensitivity. The simple invented table here keeps the first expected-count calculation completely visible.

5 Magnitude, Uncertainty, and Several Results

5.1 Magnitude and practical importance

The special-practice estimate, interval, test, and benchmark answer different parts of the substantive question. The raw difference is 3.1000 points. Its approximate interval [0.8571, 5.3429] lies above zero and agrees with the two-sided $p = .0072$ result. It also spans the course-specific two-point importance criterion.

Consequently, a positive mean assignment effect is supported under the procedure, while its practical importance remains unresolved. Effects below the criterion and effects clearly above it are both compatible with the data. The estimate exceeding two points does not settle this issue. Nor would an interval crossing zero automatically mean that all effects of practical interest had been ruled out.

A **standardized mean difference** expresses a raw difference relative to a named SD. One familiar denominator pools within-group variation:

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2}{n_1 + n_0 - 2}}, \quad d = \frac{\bar{x}_1 - \bar{x}_0}{s_{\text{pooled}}}.$$

Using the exact supplied practice summaries gives $s_{\text{pooled}} = 6.2032$ and $d = .4997$. The mean difference is about half of this particular within-group SD. Hedges' correction adjusts d for small-sample bias under its reference model; here its approximation gives $g = .4966$. Appendix B develops that correction and the regression connections.

Changing the denominator changes the comparison's scale. It does not broaden the participant scope or supply a universal importance threshold. For the current course decision, assessment points and the two-point benchmark remain the most direct terms. Chapter 4 develops effect-size interpretation across group comparisons.

5.2 Errors, power, and precision planning

A decision rule can make two kinds of error. A **Type I error** rejects a true null. A calibrated level-.05 rule limits this probability under its null conditions. A **Type II error** fails to reject under a specified alternative. If that probability is β , the rule's **power** is $1 - \beta$, its probability of rejecting under that alternative.

The possible decisions and model states form the following table. Use *do not reject*, because the decision does not reveal that the null is true.

Table 10*Decisions and error probabilities under specified states of the model.*

Decision	Null is true	Specified alternative is true
Do not reject H_0	Correct nonrejection; probability $1 - \alpha$ for an exactly calibrated continuous test	Type II error; probability β at this alternative
Reject H_0	Type I error; probability α for that calibrated test	Correct rejection; power $1 - \beta$ at this alternative

For an approximately calibrated or conservative procedure, nominal α need not equal the actual null rejection probability. β changes across alternative effects. Both error probabilities condition on a model state; neither is the probability that the decision in this completed study is wrong.

Power .80 at a particular effect means that 80% of repetitions would reject under that effect, design, and test. It leaves a .2000 chance of failing to reject there. It is neither an 80% probability that a completed study is correct nor a prediction that its exact finding will recur. The effect magnitude, outcome variability, sample size, dependence, sidedness, and decision rule all affect power. A study has a power curve across alternative effects, not one context-free power value. Appendix B uses a bounded 1,000-study illustration to distinguish the false-positive fraction among rejections from the Type I error probability.

Precision planning asks for an interval narrow enough for the intended use. Power planning asks for a stated chance of rejection at a meaningful alternative under a specified procedure. Both require inputs about anticipated variation, measurement quality, allocation, and dependence. For a class-assigned study, additional independently assigned classes can supply information that adding more students within existing classes cannot replace. A sample-size justification may use desired precision, power, feasibility, or a combination; its stated purpose explains what the study is planned to make sufficiently informative.

For a separate independent two-group study, suppose each group has anticipated SD 6.2 points and we want a Normal-approximation 95% interval with half-width 2. With equal group sizes n , the half-width is $h = z^* \sigma \sqrt{2/n}$. Solving for n gives

$$n_{\text{per group}} \approx 2 \left(\frac{z^* \sigma}{h} \right)^2 = 2 \left(\frac{1.96 \times 6.2}{2} \right)^2 = 2(6.0760)^2 = 73.8356.$$

Round upward to 74 observed outcomes per group. The calculation uses a stipulated SD and independent-sampling approximation; it does not redesign the fixed 120-participant assignment study.

The same plan illustrates why precision and power answer different questions. At 74 per group, its working SE is $6.2\sqrt{2/74} = 1.0193$. Using the full Normal critical value for a two-sided

.05 test of zero, the estimate must exceed 1.9977 or fall below -1.9977 to reject. Both tails contribute to the power calculation.

Figure 6, adapted from Biau et al. (2010), keeps that rejection rule fixed under two possible truths. At a true difference of zero, the two shaded rejection tails total $\alpha = .0500$. At a true difference of three points, the middle nonrejection area is $\beta = .1627$ and both rejection tails together give power .8373.

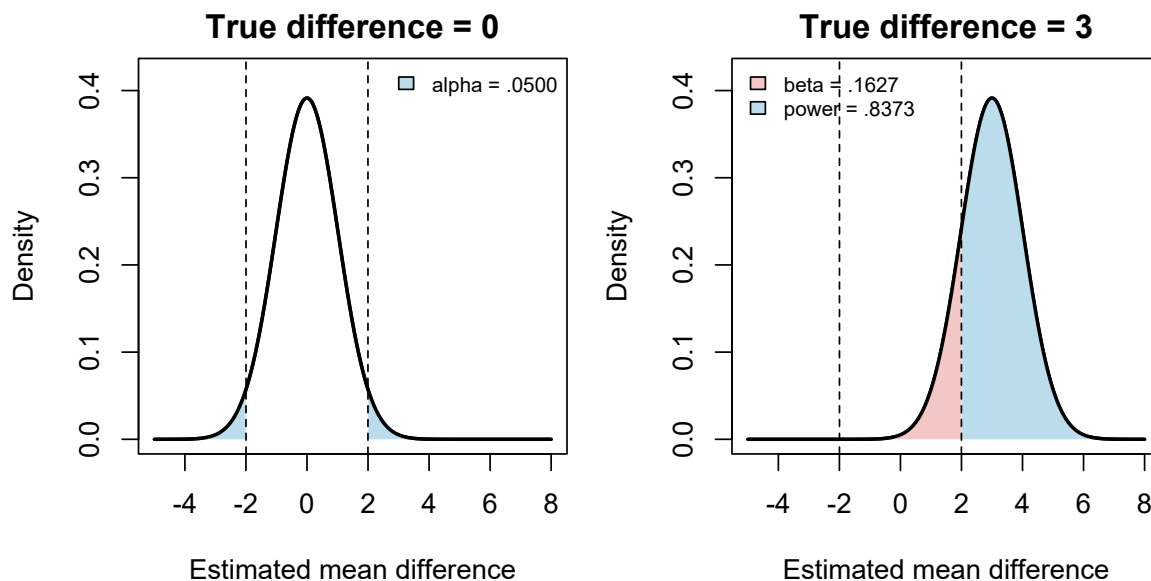


Figure 6: A fixed decision rule under two possible population differences. At zero, rejection has probability alpha; at a true three-point difference, the middle region has probability beta and both outer regions together give power. Both panels use the same rejection boundaries.

Table 11

Power for the same 74-per-group planning model.

True difference in the planning model	Rejection probability
2 points	.5009
3 points	.8373

The design chosen for a two-point interval half-width has about one-half power at a true two-point difference, rather than automatically achieving an .80 target. A larger true difference is easier to distinguish from zero with the same spread. The model, alternative, and rejection rule must accompany each power value. Appendix B gives the probability calculation.

Planning also needs a plausible spread, allocation, independently informative units, dependence structure, procedure, sidedness, threshold, measurement quality, and feasible follow-up. If 10% of outcomes are expected to be unavailable independently of relevant outcomes, $74/.90 = 82.2222$ suggests recruiting at least 83 per group to obtain 74 on average. This is no guarantee of complete data and does not repair bias from selective attrition. Unequal allocation or clustering can change the information supplied by a given number of participants.

5.3 Several comparisons and when they were chosen

One analysis can be part of a larger search. A researcher might inspect several outcomes, subgroups, models, exclusion rules, or stopping points, then report the most favorable result. The reported tail area can still be calculated, but a calibration for one advance-specified test does not automatically cover that selection process (Simmons et al., 2011). Explain which decisions were made before and which after examining the relevant results.

Advance choice and multiplicity control do different jobs. Advance choice makes the intended comparison and the scope for selection visible. Multiplicity control addresses a defined **family** of comparisons. Five planned tests can still form a family needing adjustment. A comparison noticed later can still be informative when its exploratory origin and appropriate uncertainty are reported.

The **familywise error rate** is the probability of at least one Type I error in a defined family. If five true-null tests are independent and each uses .05, the chance of no false rejection is $.95^5$. Thus

$$1 - (1 - .05)^5 = 1 - .95^5 = .2262.$$

The independence condition is essential to that calculation. For a familywise target of .05, **Bonferroni** instead allocates

$$\alpha_{\text{each}} = \frac{.05}{5} = .0100.$$

The probability of one or more false rejections is no larger than the sum of their individual probabilities. The five allowances sum to .05, so this guarantee does not require independent tests, provided the individual tests have their stated calibration. Adjustment cannot repair an invalid individual procedure or an undisclosed wider search.

Selection can also change the behavior of an otherwise unbiased estimator. Consider a Normal sampling model with true difference 1 point and known SE 2 points. A two-sided .05 test of zero using the stipulated critical value 1.96 selects estimates above 3.9200 or below -3.9200 .

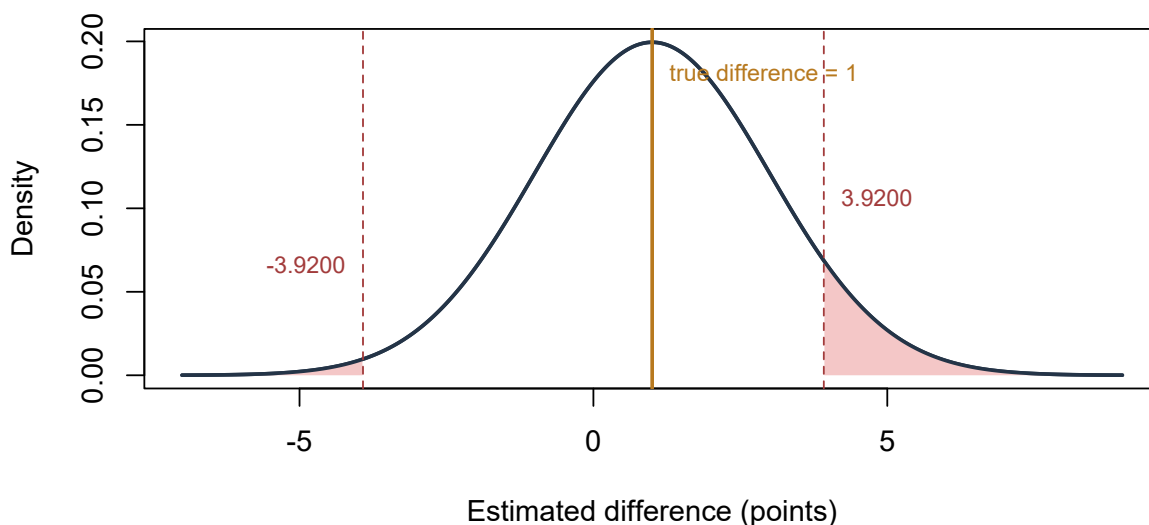


Figure 7: A significance filter retains both tails of a Normal estimate distribution centered at the true difference of one point, with known SE two points.

The selected positive estimates are much larger than the truth; the selected negative estimates have the wrong sign. The underlying estimand stays at one point, while selection changes which estimates are retained. The estimator’s original unbiasedness and its interval’s unconditional coverage need not hold within this selected set. This connects the timing of analysis choices to the comparison families developed in Chapter 4.

Chapter 4 applies this idea to intervals for several contrasts. Tukey, Dunnett, and Scheffé use other calibrations for particular families; they are not special cases of Bonferroni. The family should follow the question and the comparisons considered, rather than whichever rule produces the preferred result.

5.4 Report the result with its conditions and scope

Table 12

The invented special-practice study and its results.

Item	Special Practice	Usual Practice	Comparison or status
Assigned students	60	60	Random assignment
Delayed assessments observed	60	60	Complete follow-up
Delayed-assessment mean	27.1	24.0	3.1000-point mean difference

Item	Special Practice	Usual Practice	Comparison or status
Delayed-assessment SD	6.0	6.4	Variation among student scores
Standard error of the mean difference	—	—	1.1325 points
95% confidence interval	—	—	[0.8571, 5.3429] points
Two-sided test from the same procedure	—	—	$t(117.5119) = 2.7372$, $p = .0072$
Benchmark for potential importance	—	—	2.0 points; rationale documented before outcome access, course-specific

The completed table assembles quantities already developed. A useful results paragraph explains how they relate, rather than substituting the phrase “Special Practice worked” for the comparison:

Among the 120 consenting first-year students randomly assigned to Special Practice or Usual Practice, the mean delayed-assessment score was 3.1000 points higher under Special Practice (27.1 versus 24.0 on the 0–40 assessment), with an approximate 95% interval from 0.8571 to 5.3429 points and $p = .0072$ for the two-sided zero-effect test. The comparison and the two-point benchmark were chosen before the outcomes were seen. The interval excludes zero but spans the benchmark, supporting a positive direction while leaving practical importance unresolved. The result concerns assignment among these participants, this assessment two weeks after four sessions, and the Welch working procedure. Attendance, mechanisms, longer-term learning, and students elsewhere require their own evidence.

Random assignment supports the participant comparison, and complete follow-up avoids missing outcomes for this assessment. Its outcome and participant scope remain those stated in the report. This combination of estimates, conditions, and scope makes the result interpretable (Appelbaum et al., 2018).

Suppose the researchers now replace the primary assessment because a different one gives a larger difference. They have changed the outcome and selected it after inspection. The report should identify the original result, the new comparison, the timing and reason for the change, and the analyses considered. The new result can motivate further work; its ordinary p-value cannot be presented as if it arose from the original advance-specified procedure.

6 Reading Bridge

We have followed one estimate from its estimand and estimation rule through a stated repetition, an estimated SE, a confidence interval, and a test. Paired student scores showed why within-person differences carry the information; the school reference extends this to assignment units. The categorical transfer showed how expected counts and a reference distribution extend the same reasoning. Magnitude, practical importance, error rates, and the comparisons examined then shaped one qualified conclusion.

Chapter 4 compares several groups at once. It develops a model of group means, decomposes variation into between-group and within-group parts, and uses their mean squares to form an F ratio. Specific contrasts and their simultaneous intervals follow. A two-factor example then shows how main effects and interaction work together. The SE, interval, testing, and comparison-family ideas developed here will be used at each step.

Key terms

Use this short reference to revisit the main concepts as needed. The Part One introduction gives the broader reference and the principal home of each concept, including the sampling ideas developed in Chapter 2.

Table 13

Selected concepts for estimates, uncertainty, and tests.

Term and common alternatives	Meaning
Estimate	A numerical approximation to an estimand, calculated from the available data. A point estimate expresses the approximation as one value.
Estimator	The rule used to calculate an estimate from data. Its bias and precision concern how the rule behaves across the repetitions specified by the study's procedure or model.
Bias (of an estimator)	The difference between an estimator's expected value over the specified repetitions and the estimand. An unbiased estimator has bias zero; precision describes its spread, a separate quality.
Confidence interval (CI)	An interval expressing uncertainty about an estimand. Under its conditions, the procedure covers that estimand at its stated rate across repetitions; the rate describes the procedure.
Critical value	The cut-point of a reference distribution, such as the Normal or t, selected by a confidence or significance level. It multiplies the SE in an interval and marks the rejection region of a test.

Term and common alternatives	Meaning
Coverage	The proportion of repetitions in which an interval procedure contains the estimand under its conditions. The confidence level is the procedure's target coverage, not a probability for one realized interval.
Null hypothesis	The statistical proposition used as the reference for a test, such as equality of specified population means. The alternative hypothesis states the departure being considered.
Test statistic	A quantity calculated from data to assess departure from a null hypothesis. The test compares it with a reference distribution justified under that null and stated conditions.
One-sample z test; one-sample t test	Both compare a sample mean with a stated null mean using its SE; z uses a known process SD under its model, whereas t estimates that SD and uses the appropriate t reference. Neither choice follows an automatic sample-size cutoff.
Independent-samples t test; Welch test; pooled t test	Independent-group tests compare means. Welch allows unequal variances with an estimated reference df; the pooled procedure adds the equal-variance condition.
Paired-samples t test	A one-sample t test of within-pair differences against a specified mean difference, with independent pairs under the stated model.
p-value	Under a specified null hypothesis and conditions, the probability of a test result at least as extreme as the observed one, using the test's stated tail rule.
Significance level (alpha)	The threshold used for rejecting a null hypothesis, chosen to control the probability of rejecting a true null under the test's conditions. This error is called a Type I error.
Expected count	The count a cell of a two-way table would have under a stated null pattern such as independence, given the observed margins: row total times column total, divided by the overall total.
Pearson's chi-square statistic (X^2)	The sum over cells of squared differences between observed and expected counts, each divided by its expected count, compared with a chi-square reference with $(R - 1)(C - 1)$ degrees of freedom under the stated model.
Power (statistical power)	The probability of rejecting the null under a specified alternative, design, and test. Failure to reject under that alternative is a Type II error; power is one minus its probability.
Familywise error rate (FWER)	The probability of at least one Type I error among a defined family of tests. Procedures such as Bonferroni allocate the error allowance across that family.

Further reading

The GAISE College Report emphasizes statistical investigation, conceptual understanding, and interpretation (American Statistical Association, 2016). *Introduction to Modern Statistics* develops inference through repetition, with separate treatments of paired means and two-way tables (Çetinkaya-Rundel & Hardin, 2024). Penn State STAT 200, Lesson 4 connects interval construction with its interpretation. The ASA p-value statement (Wasserstein & Lazar, 2016) and Moving to a World Beyond $p < .05$ explain why uncertainty and context deserve more attention than a threshold label.

Exercises

These exercises use the information supplied and a standard calculator. They practise calculation, interpretation, and the conditions behind a conclusion.

Exercise 1: Name the inferential objects (Foundational)

A report states:

The estimand was the mean Special-minus-Usual assignment contrast in delayed-assessment score among the 120 randomized participants. In every admissible repetition, the analyst would subtract the Usual assigned-group mean from the Special assigned-group mean. The observed difference was 3.1000 points, with an estimated standard error of 1.1325 points.

Identify:

1. the estimand;
2. the estimator;
3. the estimation process;
4. the estimate and one statistic used in obtaining it;
5. the standard error and what varies in its repeated-process interpretation;
6. the parameter; and
7. two bounds that must remain in the answer.

Explain why the estimate and the parameter should not be given the same label.

Exercise 2: Calculate SE and distinguish it from SD (Foundational)

The Special and Usual Practice groups each contain 60 students. Their respective delayed-score SDs are 6.0 and 6.4 points. The reported uncertainty calculation is

$$\widehat{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

1. Substitute the group sizes and SDs to check the reported SE of 1.1325 points.
2. Which quantities describe variation among scores, and which describes estimated variation of the mean difference across possible assignments?
3. Why does the SE retain assessment-point units?
4. If effective information increased while the design and score variation remained comparable, would the SD or SE usually become smaller?
5. Name two defects that a smaller standard error would not repair.
6. Does random assignment make the Welch reference an exact randomization distribution? Explain briefly.

Exercise 3: Interpret interval coverage (Foundational)

A student writes:

The 95% confidence interval [0.8571, 5.3429] means there is a 95% probability that the true mean assignment effect lies between 0.8571 and 5.3429 points. If the method is valid, every 95% interval should contain the true effect.

Rewrite the statement using repeated-procedure language. Then explain:

1. why some intervals from a valid 95% procedure miss the fixed estimand;
2. what the endpoints contribute to reading this study; and
3. three sources of uncertainty or bias that the conventional interval may omit.

Exercise 4: Translate a p-value (Foundational)

For the matching two-sided zero-contrast test, $t(117.5119) = 2.7372$ and $p = .0072$. Consider four interpretations:

- A. There is a 0.72% probability that the null hypothesis is true.
- B. There is a 99.28% probability that Special Practice will work again.
- C. Under the zero-contrast null model and the stated procedure, results at least as incompatible as the observed one have probability about .0072.
- D. The 3.1000-point effect is educationally important because $p = .0072$.

Choose the defensible interpretation and diagnose each rejected one. Then use an estimate of 3.1, an SE of 1.13255, and a supplied 95% critical value of 1.98036 to:

1. check $t = (3.1 - 0)/1.13255$;
2. calculate $3.1 \pm 1.98036(1.13255)$ and round the endpoints to four decimal places;
3. explain why exclusion of zero agrees with the matching two-sided $p < .05$ result, without implying exact randomization calibration; and
4. write one sentence reporting estimate, interval, and p-value without using **prove**, **accept**, or **worked**.

The reference tail area is supplied as $p = .0072$; you do not need to calculate a t -distribution probability by hand.

Exercise 5: Interpret one published active-learning result (Integrative)

Deslauriers and colleagues (2019) used a randomized crossover study in introductory college physics: students were randomly assigned to two groups, and each group experienced both instructional conditions in successive class meetings, in opposite orders. In a regression model, performance on a test of learning was 0.46 SD higher under active instruction than under passive instruction, $p < .001$. The brief result supplied here does not report a confidence interval (Deslauriers et al., 2019).

Use the supplied description to:

1. mark the participant set, condition contrast, outcome, estimate, units, p-value, and design;
2. explain what the estimate and p-value each tell you;
3. identify the information supplied and what else would help you understand the result; and
4. state a bounded one-sentence version of the reported answer.

Then explain which design and analysis details would help judge the uncertainty calculation and scope. State what further information you would seek in the Methods or supplement and name one broader claim this excerpt cannot support.

Exercise 6: Repair the special-practice conclusion (Integrative)

Rewrite:

Special Practice worked, $p < .05$.

Use this study record (the same one summarized in the chapter's study-and-results table): 120 consenting students in one course were randomly assigned, 60 per group, to four sessions of Special or Usual Practice with equal total study time. All completed a common 0–40 assessment

two weeks later. Special had mean 27.1 (SD 6.0), and Usual had mean 24.0 (SD 6.4). The prespecified primary comparison gave 3.1000 points, 95% CI [0.8571, 5.3429], approximate Welch $t(117.5119) = 2.7372$, $p = .0072$. Before outcome access, the protocol specified the two-point benchmark as potentially important: the course team treated two points as the smallest average difference that would change its recommendation, because the marking scheme awards two points to one completed application task and the new format would add little preparation time. Your revision must retain:

1. the participant set and assignment comparison;
2. the outcome, time, direction, and raw units;
3. the estimate and 95% interval;
4. the reported numerical p-value only after the estimate and interval;
5. the choice of the primary comparison before outcome access;
6. what random assignment and complete follow-up support;
7. the interval's relation to the prespecified, course-specific two-point benchmark for potential importance; and
8. at least three scope boundaries.

After the rewrite, choose three phrases and explain what each repairs in the original sentence.

After viewing the outcomes, the researchers decide that a different assessment should replace the planned assessment as primary because it gives a larger difference. Explain how this changes the comparison and its interpretation. State what information should be reported about the original and new analyses, what advance-planning claim can no longer be made, and why the new assessment's p-value cannot be read as if that assessment had been chosen before inspection.

Exercise 7: Check expected counts and a chi-square result (Foundational)

An invented study records participation format and completion. It is distinct from the UCI student data.

Table 14

Participation-format outcomes for Exercise 7.

Participation format	Completed	Did not complete	Total
Online	84	36	120
In person	63	17	80
Total	147	53	200

The expected counts under independence are 88.2000, 31.8000, 58.8000, and 21.2000. The Online/Completed contribution is $(84 - 88.2000)^2 / 88.2000 = 0.2000$; the four contributions (0.2000, 0.5547, 0.3000, 0.8321) sum to $X^2 = 1.8868$ on $(2 - 1)(2 - 1) = 1$ df, giving $p = .1696$.

Report calculated results to four decimal places. The completion proportions are $84/120 = .7000$ and $63/80 = .7875$.

1. Reproduce the Online/Completed expected count, its contribution, and the degrees of freedom, explaining what an expected count represents.
2. Identify the cells in which the observed count exceeds the expected count.
3. State the omnibus null in words.
4. Write one conclusion that distinguishes the observed proportion difference, evidence against independence, association magnitude, and causation.
5. Explain why $p = .1696$ is not proof that format and completion are independent.

Appendix A: Probability and Repetition Foundations

This appendix develops the probability and variance results behind the chapter's uncertainty calculations. Each result belongs to a stated repetition: drawing observations from a model, sampling a finite list, or reassigning the same participants. Keeping that repetition visible explains both the usefulness of a formula and the limits of its application.

Probability, models and expectation

Chapter 2 introduces conditional probability, independence and the distribution catalogue. Recall the distinction needed here. An empirical distribution describes observed values; a probability model assigns chances to possible values; a sampling distribution describes a statistic over the stated repetitions. A conditional probability $P(A | B) = P(A \cap B)/P(B)$ uses only outcomes satisfying B . Independence of two events means $P(A \cap B) = P(A)P(B)$; independence of observations is a separate condition in the SE calculations.

Chapter 2 defined the expectation of a discrete random variable as the probability-weighted mean $E(X) = \sum_x xP(X = x)$; for a continuous variable an integral replaces the sum. For example, the Binomial model with ten independent trials and constant success probability .70 has expected count $10(.70) = 7.0000$ and variance $10(.70)(.30) = 2.1000$. Seven is an average over the model's repetitions, not a prediction for every ten trials. We now use expectation and covariance to derive uncertainty.

Variance, sample variance, and averages

The variance of a random variable is its expected squared distance from its expectation: $\text{Var}(Y) = E\{(Y - E(Y))^2\}$. For fixed weights a_i and observations with finite second moments,

$$\text{Var} \left(\sum_i a_i Y_i \right) = \sum_i a_i^2 \text{Var}(Y_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(Y_i, Y_j).$$

Here covariance is the expected product of two centered observations, and $i < j$ includes each distinct pair once. Expanding the square of the weighted sum gives the squared terms and the cross-products in this identity. For an average, every weight is $1/n$, giving

$$\text{Var}(\bar{Y}) = \frac{1}{n^2} \left\{ \sum_i \text{Var}(Y_i) + 2 \sum_{i < j} \text{Cov}(Y_i, Y_j) \right\}.$$

Independent observations have zero covariance. If their common variance is σ^2 , the expression becomes $n\sigma^2/n^2 = \sigma^2/n$. For a difference, weights 1 and -1 instead give $\text{Var}(U - V) = \text{Var}(U) + \text{Var}(V) - 2 \text{Cov}(U, V)$. These two applications explain the mean, independent-group, and paired-difference calculations in the chapter.

Estimating σ^2 introduces the sample variance, $s^2 = \sum_i (Y_i - \bar{Y})^2 / (n - 1)$. The deviations satisfy $\sum_i (Y_i - \bar{Y}) = 0$: once $n - 1$ deviations are specified, the last is fixed. This constraint helps explain the denominator, but its unbiasedness requires a calculation. For independent observations with common mean μ and variance σ^2 ,

$$\sum_i (Y_i - \bar{Y})^2 = \sum_i (Y_i - \mu)^2 - n(\bar{Y} - \mu)^2.$$

The expectation of the first term on the right is $n\sigma^2$. The sample mean is unbiased and has variance σ^2/n , so the second term has expectation $n(\sigma^2/n) = \sigma^2$. Consequently,

$$E(s^2) = \frac{n\sigma^2 - \sigma^2}{n - 1} = \sigma^2.$$

For the three observed values 2, 4 and 6, the mean is 4 and the squared deviations sum to 8. The sample variance is $8/(3 - 1) = 4.0000$, whereas the squared-deviation average $8/3 = 2.6667$ uses a different denominator.

Normality is unnecessary for this expectation result; the independence, common mean and variance, and finite-variance conditions matter. Unbiased estimation of the variance does not imply that its square root s is unbiased for σ .

The observed-data version also prepares Chapter 4's sums of squares. For any proposed center a , $\sum_i (y_i - a)^2 = \sum_i (y_i - \bar{y})^2 + n(\bar{y} - a)^2$. The last term is nonnegative and vanishes at $a = \bar{y}$, so the mean minimizes the sum of squared deviations. This **least-squares** property is an algebraic fact about the observed values.

Concentration and limiting sampling shape

The running-mean illustration in Section 2.4 separates concentration of one accumulating sequence from the sampling-distribution shape across repetitions. The formal statements make their conditions explicit.

For independent, identically distributed observations with finite absolute mean, $E(|Y|) < \infty$, the **law of large numbers** states

$$\bar{Y}_n \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty.$$

Convergence in probability means that, for every fixed positive tolerance ε , $P(|\bar{Y}_n - \mu| > \varepsilon)$ tends to zero. It describes concentration near the mean, rather than a requirement that every successive average move closer to it.

For independent, identically distributed observations with finite, nonzero variance σ^2 , the **central limit theorem** states

$$\frac{\sqrt{n}(\bar{Y}_n - \mu)}{\sigma} = \frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Convergence in distribution means that the centered mean, measured in its own SE units, approaches a standard Normal distribution. The theorem concerns the sampling shape of the mean, not the distribution of individual waiting times. Strong skewness or heavy tails can slow this approximation; dependence may require a different theorem and SE. A universal sample-size cutoff cannot certify the approximation. Neither limiting result repairs selection, measurement, missing-data, or design problems.

Dependence, finite populations, and effective information

The covariance terms in the variance identities show how dependence changes precision. Positive total covariance makes a mean more variable than the independent-observation formula suggests. Negative total covariance can make it less variable. Independence implies zero covariance when second moments exist; zero covariance alone allows nonlinear dependence.

Chapter 2's correlation-bound calculation shows why a standardized covariance lies between -1 and 1 . Here the covariance terms matter because they change the uncertainty of a mean or a difference.

Sampling without replacement from a fixed list produces dependence even when selection is random. For simple random sampling without replacement from N fixed population values, let $\bar{Y}_N$ be the population mean and define the population variance using divisor N :

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y}_N)^2, \quad SE(\bar{Y}) = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}.$$

The **finite-population correction**, $\sqrt{(N-n)/(N-1)}$, accounts for the decreasing number of unobserved units; the sampling uncertainty is zero at a census. Other selection schemes introduced in Chapter 2 require their own design-based variance calculation. Sampling without replacement alone does not establish this formula.

An equivalent convention uses S_N^2 , the same population squared deviations divided by $N-1$. Since $S_N^2 = N\sigma^2/(N-1)$, it gives

$$SE(\bar{Y}) = \sqrt{1 - \frac{n}{N}} \frac{S_N}{\sqrt{n}}.$$

With this convention, the correction for 100 of 20,000 units is $\sqrt{1 - .0050} \approx .9975$; with 400 it is $\sqrt{1 - .0200} \approx .9899$. The choice of population-variance divisor explains the different-looking factors; both formulas give the same SE. This correction for the sampling fraction serves a different purpose from the sample-variance denominator $n-1$.

A **precision-equivalent effective sample size** expresses a comparison with a specified independent-sampling benchmark. If that benchmark gives variance σ^2/n , define n_{eff} by $\text{Var}_{\text{design}}(\bar{Y}) = \sigma^2/n_{\text{eff}}$. For equal clusters of size m , equal weights, common marginal variance, within-cluster correlation ρ , and independent clusters, $n_{\text{eff}} = n/[1 + (m-1)\rho]$. With $m = 20$ and $\rho = .05$, the denominator is 1.9500, so these observations supply about half the mean-estimation precision of equally many independent observations. Different weighting or dependence changes the comparison; an efficient design can yield $n_{\text{eff}} > n$. This quantity is specific to the estimand and variance benchmark. It is neither the number of assignment units nor a universal replacement for degrees of freedom.

Random assignment and reference calibration

For the fixed $N = 120$ Special Practice participants, let S_1^2 and S_0^2 be the potential-outcome variances under the two assignments, with divisor $N-1$. Let S_{eff}^2 be the variance of the individual effects $Y_i(1) - Y_i(0)$, using the same divisor. Under complete random assignment of n_1 and n_0 participants, consistent conditions and no relevant interference (Bai et al., 2024),

$$\text{Var}_{\text{assign}}(\bar{Y}_1 - \bar{Y}_0) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_{\text{eff}}^2}{N}.$$

The assigned groups are complementary, so their means need not be independent. Only one potential outcome is observed per participant, leaving the individual-effect variance unidentified.

The observed within-group variances estimate the corresponding potential-outcome variances. Consequently,

$$E_{\text{assign}} \left(\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0} \right) = \text{Var}_{\text{assign}}(\bar{Y}_1 - \bar{Y}_0) + \frac{S_{\text{eff}}^2}{N}.$$

The familiar summed variance is conservative in expectation under this design, with equality when individual effects are constant. A particular estimate need not be too large. This expectation identity alone neither makes a Welch t reference exact nor guarantees conservative finite-sample interval coverage. For genuinely independent samples the same SE formula has another justification: their mean variances add with zero covariance.

The **Welch–Satterthwaite degrees of freedom** are

$$\nu = \frac{(s_1^2/n_1 + s_0^2/n_0)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_0^2/n_0)^2/(n_0 - 1)}.$$

Using 36/60 and 40.96/60 gives $\nu = 117.5119$ and critical value 1.9804. Fractional df describe the reference approximation, not fractional people. Welch’s independent-sample procedure allows unequal variances (Welch, 1947); its application to the participant assignment remains the chapter’s working approximation.

A zero average effect allows positive and negative individual effects to cancel. The stronger **sharp null** says every participant’s effect is zero. A randomization test of that stronger proposition uses the actual assignment mechanism, rather than obtaining exactness from a Normal or t reference. The cited randomized-experiment reading develops its full derivation; the distinction is enough to interpret this chapter’s working mean comparison (Bai et al., 2024).

For a formal coverage statement, write $C(D)$ for the interval from a possible data record D . Then $P\{\theta \in C(D)\}$ concerns a fixed estimand across the declared repetitions. Exact 95% coverage equals .95 under its model, approximate coverage approaches that target under justified conditions, and conservative coverage is at least .95. Other quantities, such as an unknown variance, can affect this probability. The model and design therefore accompany the coverage claim; it assigns no probability to the fixed estimand after observing one interval.

Bias, variance, and mean squared error

Mean squared error combines an estimator’s systematic displacement and its variability under one stated repetition:

$$\text{MSE}(\hat{\theta}) = E\{(\hat{\theta} - \theta)^2\} = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

To see the identity, write the estimation error as $\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta$ and square it. The first part has expectation zero, so the expected cross-product vanishes. The remaining terms are the variance and squared bias.

An unbiased estimator with SD 2 has $\text{MSE } 2^2 + 0^2 = 4.0000$. An estimator with SD 1 and bias 1.5 has $\text{MSE } 1^2 + 1.5^2 = 3.2500$. The second has lower average squared error in this comparison, despite its bias. That judgment depends on the common estimand, repetition, and squared-error criterion. Precision alone cannot establish accuracy, and a favorable MSE under a model cannot repair an unsuitable outcome or target population.

Appendix B: Inferential Extensions

These extensions connect the chapter's worked calculations to related procedures and planning choices. The target quantity and the independent or assigned units determine the relevant formula. Chapter 4 develops the comparison of several means and comparison families in greater detail.

A compact guide to common procedures

The named procedures in Section 4.1 connect the observed units to their SE and reference. Here we connect their formulas and distinguish estimation from a null restriction. Many intervals have the form $\hat{\theta} \pm c\widehat{SE}(\hat{\theta})$, where c is a critical value. Many test statistics have the form $(\hat{\theta} - \theta_0)/SE_0(\hat{\theta})$, where the subscript zero reminds us that the SE may be calculated under a null restriction. The two expressions match only when their estimators, restrictions, reference distributions, and sidedness are compatible.

Under the independent Normal model for one mean,

$$\bar{Y} \pm t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}}, \quad t = \frac{\bar{Y} - \mu_0}{s/\sqrt{n}}.$$

For n independent pairs, apply these formulas to the pair differences: replace $\bar{Y}$ by $\bar{D}$, s by s_D , and μ_0 by $\mu_{D,0}$. Normality for an exact t reference concerns the differences. For two independent groups, use the sum of the two mean variances and the Welch degrees of freedom in Appendix A. For a population SD known independently of the sample, a suitable Normal model instead gives a z reference for a mean; such knowledge is unusual in applied social research.

For n independent binary observations with a common probability p , $SE(\hat{p}) = \sqrt{p(1-p)/n}$. Estimation commonly replaces p by $\hat{p}$; a test of $p = p_0$ may use p_0 in its null SE. A correct variance formula alone does not ensure that a Normal interval has reliable coverage for small or boundary proportions.

Sidedness also changes the procedure. For a continuous symmetric t or z reference, let p_2 be the usual two-sided p-value. If the statistic points in an alternative direction chosen before inspecting the relevant results, its one-sided p-value is $p_2/2$. If it points in the opposite direction, it is $1 - p_2/2$. The special-practice statistic therefore gives approximately .0036 for a positive alternative and .9964 for a negative alternative; its reported procedure remains two-sided. This halving shortcut need not hold for asymmetric or discrete tests. A one-sided level-.05 test matches a one-sided 95% confidence bound, with a different cutoff from a two-sided 95% interval.

Resampling and uncertainty

A **bootstrap distribution** approximates an estimator's variability by repeatedly resampling from observed data. Its center is usually near the observed estimate. In a simple case bootstrap for independent observations:

1. Identify the independent observational unit and the original number of units.
2. Draw that many units with replacement from the observed units.
3. Apply the same estimator to the resampled data.
4. Repeat the resampling and estimation many times.
5. Use the distribution of estimates to examine variability or construct an appropriate interval.

The bootstrap SE is the sample SD of the resulting estimates. It is distinct from the SD of individual observations and from the uncertainty caused by using only finitely many simulation repetitions. Preserve the study structure: resample pairs together, or clusters when independent clusters are the appropriate units. Resampling individual students from clustered classes can misrepresent uncertainty. The number resampled in the simple algorithm is the observed unit count, not a substituted precision-equivalent effective sample size.

A **null distribution** instead describes a test statistic under specified null restrictions and other conditions. A centered t statistic may have a zero-centered null reference; a chi-square or F statistic does not. A bootstrap distribution centered near an observed difference is therefore not automatically a null distribution for testing zero. A randomization test must reproduce the allowed assignment mechanism under the null being tested. Resampling can represent an appropriate repetition; it cannot recover unobserved cases or repair biased selection, invalid measurement, or an unsuitable estimand.

Paired schools: a comparison of class means

Each of twelve participating schools contributes two classes of twenty students. Within each school, one class is randomly assigned to explanatory feedback and the other to usual feedback. All 480 students complete the same delayed 0–40 assessment. These are invented records; the

schools are participating schools rather than a probability sample of all schools. The class is the assignment unit, and each school supplies one explanatory-minus-usual contrast.

Table 15

Invented class means under within-school assignment.

School	Usual mean	Explanatory mean	Difference
A	21	19	-2
B	22	22	0
C	20	21	1
D	23	24	1
E	24	26	2
F	22	24	2
G	25	28	3
H	24	27	3
I	23	27	4
J	26	30	4
K	25	30	5
L	22	29	7

The twelve differences sum to 30. Weighting schools equally gives $\bar{d} = 30/12 = 2.5000$. Equal class sizes also make student weights equal in this example. The squared deviations from 2.5 sum to 63:

$$s_d = \sqrt{63/11} = 2.3932, \quad \widehat{SE}(\bar{d}) = \frac{\sqrt{63/11}}{\sqrt{12}} = 0.6908.$$

The zero-mean contrast test uses $t = 2.5/(\sqrt{63/11}/\sqrt{12}) = 3.6187$, with 11 df and $p = .0040$. The interval is

$$\bar{d} \pm t_{.975,11} \widehat{SE}(\bar{d}) = 2.5000 \pm 2.2009852 \frac{\sqrt{63/11}}{\sqrt{12}} \approx [0.9795, 4.0205] \text{ points.}$$

This approximate t calibration represents variation among the twelve school contrasts. Implemented within-school assignment, the common outcome and follow-up, and no relevant interference between classes support the interpretation. It does not make these schools representative of a wider population. Replacing 12 by 480 in the SE denominator would discard the assignment design.

This is a **blocked assignment**: each school forms a block, and the two conditions are allocated within every school. Comparisons then use the within-school differences. Blocking can remove school-level variation from those contrasts. **Stratified sampling** instead selects observations

within population strata to determine who enters a sample. It addresses selection into the study; blocking here addresses allocation after entry. The similar use of groups does not make either mechanism imply the other. Chapter 4 recalls this design when discussing ANOVA conditions.

Error probabilities across 1,000 studies

Consider stipulated expected counts across many studies, not empirical counts about education research. Suppose 900 of 1,000 studies have a true null and 100 have the specified alternative. Every test has $\alpha = .05$ under its null model and power .80 at that alternative.

Table 16

Expected decisions under stipulated prevalence, alpha and power.

Model state	Reject	Do not reject	Total
True null	45	855	900
Specified alternative	80	20	100
Total	125	875	1,000

The null-state Type I error probability is $45/900 = .0500$; at the alternative the Type II error probability is $20/100 = .2000$. Among the expected 125 rejections, however, $45/125 = .3600$ arise from true nulls. This false-positive fraction has a different denominator. Changing the stipulated frequency of true nulls or the power changes it. It is neither an observed p-value nor a probability to assign directly to the null in one completed study.

Standardized effects and regression connections

A standardized mean difference expresses a raw difference relative to a named spread. For two independent groups, one familiar choice is

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2}{n_1 + n_0 - 2}},$$

$$d = \frac{\bar{y}_1 - \bar{y}_0}{s_{\text{pooled}}}, \quad g \approx d \left(1 - \frac{3}{4(n_1 + n_0) - 9} \right).$$

The pooled SD combines the within-group sums of squares. Hedges' g applies a small-sample bias correction to d ; the displayed multiplier approximates the usual correction under its reference model (Lakens, 2013). The special-practice summaries give $s_{\text{pooled}} = \sqrt{38.4800} \approx 6.2032$, $d = .4997$, and $g \approx .4966$. These scale summaries accompany the 3.1000-point raw difference; they do not change its participant scope or certify its educational importance.

A within-person standardized change could use the SD of change scores, the pretest SD, or another reference spread. These denominators express different comparisons. Standardization can assist comparison across measures, but it removes the original unit, and conventional size labels are not universal importance criteria. Chapter 4 applies effect-size reasoning to comparisons among several groups.

Several familiar estimates also have regression representations. A one-sample mean is the fitted intercept in an intercept-only model; the mean paired change is the intercept when the outcome is the within-pair difference. With $X_i = 1$ for Special assignment and 0 for Usual,

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

fits the Usual mean as $\hat{\beta}_0$ and the Special-minus-Usual difference as $\hat{\beta}_1$. This is an algebraic equality of ordinary-least-squares coefficients. Uncertainty still requires the appropriate variance estimator and design; the ordinary equal-variance regression SE is not automatically the Welch SE. The representation does not assert a constant individual treatment effect or create a population estimand. Later chapters develop the additional assumptions and uses of regression.

Power, precision, selection, and several comparisons

Section 5.2 develops the planning inputs, the 74-per-group example, and the distinction between interval width and power. This appendix supplies the algebra, probability calculation, and further consequences.

Precision planning begins with a target interval width in meaningful units. In a separate model with two independent, equally sized groups, common planning SD σ , and a Normal critical value z^* , the SE of the difference is $\sigma\sqrt{2/n}$. Setting the desired half-width to $h = z^*\sigma\sqrt{2/n}$ and solving for n gives

$$n_{\text{per group}} \approx 2 \left(\frac{z^*\sigma}{h} \right)^2 = 2 \left(\frac{1.96(6.2)}{2} \right)^2 = 73.8356.$$

Rounding upward gives 74 observed outcomes per group. A finite-t critical value, estimated rather than known spread, unequal allocation, or clustering can change the requirement. The main discussion explains the attrition allowance. Halving the target half-width approximately quadruples the required sample size under this model.

For the power comparison, write the estimator as $\widehat{\Delta\mu} \sim N(\Delta\mu, SE^2)$, where $SE = 6.2\sqrt{2/74}$, and let $c = z_{.975}SE$. The two-sided test rejects when the estimate is below $-c$ or above c . With Φ denoting the standard Normal cumulative distribution function,

$$\text{Power}(\Delta\mu) = \Phi\left(\frac{-c - \Delta\mu}{SE}\right) + 1 - \Phi\left(\frac{c - \Delta\mu}{SE}\right).$$

Using the unrounded SE and $z_{.975}$ gives $c = 1.9977$ and the two powers in Section 5.2. At $\Delta\mu = 0$, this expression is the .05 Type I error probability. A power curve repeats the calculation across plausible alternatives, retaining the same design and test.

The significance filter in Section 5.3 changes the distribution being summarized. If S denotes selection, the conditional mean $E(\widehat{\Delta\mu} \mid S)$ need not equal $E(\widehat{\Delta\mu})$, and $P\{\Delta\mu \in C(D) \mid S\}$ need not equal the interval's unconditional coverage. Selection affects what is retained without changing the underlying estimand.

The relevant choices can extend beyond the tests in a reported table: outcomes, time points, subgroups, transformations, coding rules, exclusions, model specifications, repeated looks, and stopping decisions all provide opportunities for selection (Simmons et al., 2011). A useful report identifies which choices preceded inspection, which followed it, and which results are available. Exploratory findings can motivate further study, while their interpretation should preserve how they were found.

For m tests with valid individual null p-values, Bonferroni's rule uses α/m per test. The union bound gives

$$P(\text{at least one false rejection}) \leq \sum_{j \in \mathcal{H}_0} P(\text{false rejection in test } j) \leq m \frac{\alpha}{m} = \alpha,$$

where $\mathcal{H}_0$ indexes the null hypotheses that are actually true. The bound allows dependent tests and also applies when some nulls are false. Five tests with a familywise target .05 therefore use .0100 each. By contrast, five *independent*, true-null tests each conducted at .05 have probability $1 - .95^5 = .2262$ of at least one false rejection. Dependence changes that product calculation.

The guarantee concerns the declared family and valid component procedures; it cannot automatically cover an undocumented search. For corresponding two-sided intervals, Bonferroni uses confidence level $1 - \alpha/m$ for each member to obtain at least $1 - \alpha$ simultaneous coverage. Chapter 4 applies this idea to specific contrasts and develops other procedures suited to different comparison families. Tukey, Dunnett, and Scheffé procedures have their own reference calculations and are not special cases of Bonferroni.

References

- American Statistical Association. (2016). *Guidelines for assessment and instruction in statistics education (GAISE) college report 2016*. American Statistical Association. https://www.amstat.org/asa/files/pdfs/GAISE/GaiseCollege_Full.pdf
- Appelbaum, M., Cooper, H., Kline, R. B., Mayo-Wilson, E., Nezu, A. M., & Rao, S. M. (2018). Journal article reporting standards for quantitative research in psychology: The APA publications and communications board task force report. *American Psychologist*, 73(1), 3–25. <https://doi.org/10.1037/amp0000191>
- Bai, Y., Shaikh, A. M., & Tabord-Meehan, M. (2024). *A primer on the analysis of randomized experiments and a survey of some recent advances*. arXiv preprint arXiv:2405.03910. <https://home.uchicago.edu/~amshaikh/webfiles/jpemicreview.pdf>
- Biau, D. J., Jolles, B. M., & Porcher, R. (2010). P value and the theory of hypothesis testing: An explanation for new researchers. *Clinical Orthopaedics and Related Research*, 468(3), 885–892. <https://doi.org/10.1007/s11999-009-1164-4>
- Çetinkaya-Rundel, M., & Hardin, J. (2024). *Introduction to modern statistics* (2nd ed.). OpenIntro. <https://openintro.org/book/ims/>
- Deslauriers, L., McCarty, L. S., Miller, K., Callaghan, K., & Kestin, G. (2019). Measuring actual learning versus feeling of learning in response to being actively engaged in the classroom. *Proceedings of the National Academy of Sciences*, 116(39), 19251–19257. <https://doi.org/10.1073/pnas.1821936116>
- Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4, 863. <https://doi.org/10.3389/fpsyg.2013.00863>
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11), 1359–1366. <https://doi.org/10.1177/0956797611417632>
- Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2), 129–133. <https://doi.org/10.1080/00031305.2016.1154108>
- Welch, B. L. (1947). The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, 34(1–2), 28–35. <https://doi.org/10.1093/biomet/34.1-2.28>